

E083

Mathematics of AI

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A vision model trained on a cluster of 8 GPUs has a weight $w = 2.0$. Its dataset has 1.2 million images. The gradient $\partial L / \partial w$ is 0.5 and the learning rate is 0.1. What is w after one gradient descent step?

- A. 1.95
- B. 2.05
- C. 1.5
- D. 0.5

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

One gradient descent step with learning rate 0.2 moved a weight from 3.0 to 2.6. What was the gradient?

- A. 0.08
- B. 2.0
- C. 0.4
- D. -2.0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A speech team at a company with offices in four time zones sees training loss zigzag and climb higher each epoch. They use learning rate 0.5 and a batch size of 256 on a new cloud cluster. What change should they try first?

- A. Add more layers to the network.
- B. Raise the learning rate to escape faster.
- C. Lower the learning rate.
- D. Stop training; rising loss always means overfitting.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

In a small network built for a university research project, $y = \sigma(z)$ with $z = w \cdot x$. For one input from a 10,000-row dataset, $x = 3$ and $\sigma(z) = 0.8$. What is $\partial y / \partial w$?

- A. 0.16
- B. 2.4
- C. 0.48
- D. -0.48

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A churn model at a telecom with 9 million subscribers scores 98% on training data and 81% on validation. The team, which recently moved to a new building, increases the L2 penalty λ . What is the most likely result?

- A. All error disappears on both sets.
- B. Variance increases, widening the gap.
- C. Training accuracy rises above 98%.
- D. Training accuracy dips a little and the train-validation gap narrows.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A factory's visual inspection model, sold by a vendor with glowing reviews, catches 90% of defective parts and falsely flags 5% of good parts. Only 2% of parts are defective. The factory runs three shifts. If a part is flagged, about how likely is it truly defective?

- A. 90%
- B. About 27%
- C. 2%
- D. 95%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Using the same numbers (2% defective, 90% caught, 5% of good parts falsely flagged), what share of all parts gets flagged?

- A.** 1.8%
- B.** 4.9%
- C.** 95%
- D.** About 6.7%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A classifier for a wildlife nonprofit with volunteers on three continents assigns probability 0.25 to the correct species and 0.75 to the others combined. The photo was taken at dawn. Using the natural log, what is the cross-entropy loss for this example?

- A.** 2.0
- B.** 0.25
- C.** About 1.39
- D.** About 0.29

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

One example's cross-entropy loss (natural log) is 0.105. About what probability did the model give the true class?

- A.** About 0.90
- B.** 0.105
- C.** About 0.10
- D.** About 1.11

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A 12-layer network using sigmoid activations, trained overnight in a lab with a view of the harbor, shows gradients at the first layer about one-millionth the size of those at the last. The lab just bought new chairs. What is the best fix?

- A. Add more sigmoid layers.
- B. Switch to ReLU-type activations and add residual connections.
- C. Nothing; vanishing gradients only affect the output layer.
- D. Multiply the learning rate by 1,000.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A fraud model at an online marketplace, which just launched in Brazil, scores 80% accuracy on a test set of 2,500 examples. The test set was labeled by contractors. What is the standard error of that accuracy estimate?

- A. 0.8 percentage points
- B. 20 percentage points
- C. 0.0064%
- D. About 1.6 percentage points

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

With 1,000 test examples, the margin of error is ± 3 points. About how many examples are needed for ± 1.5 points?

- A. About 1,414
- B. 500
- C. 2,000
- D. 4,000

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A vision model ~~trained on a cluster of 8 GPUs~~ has a weight $w = 2.0$. ~~Its dataset has 1.2 million images.~~ The gradient $\partial L / \partial w$ is 0.5 and the learning rate is 0.1. What is w after one gradient descent step?

Target word: updated weight **Decides:** $w - \eta \cdot g = 2.0 - 0.1 \times 0.5$.

A	Correct	$2.0 - 0.05 = 1.95$.
B	Inverted operation	Adds the gradient step, which climbs the loss.
C	Dropped element	Leaves out the learning rate.
D	Wrong stem number	The gradient given in the stem, not the new weight.

2. Answer B Concept 1 · transfer

One gradient descent step with learning rate 0.2 moved a weight from 3.0 to 2.6. What was the gradient?

Target word: solve for the gradient **Decides:** $g = (w_{\text{old}} - w_{\text{new}}) / \eta = 0.4 / 0.2$.

A	Adjacent category	Multiplies the change by η instead of dividing.
B	Correct	$0.4 / 0.2 = 2.0$.
C	Dropped element	The change in weight, before dividing by η .
D	Inverted operation	A negative gradient would have increased the weight.

3. Answer C Concept 2

A speech team ~~at a company with offices in four time zones~~ sees training loss zigzag and climb higher each epoch. They use learning rate 0.5 ~~and a batch size of 256 on a new cloud cluster~~. What change should they try first?

Target word: fix diverging loss **Decides:** Oscillating, growing loss signals steps that overshoot.

A	Adjacent category	Capacity does not cure an unstable step size.
B	Inverted operation	Larger steps make overshooting worse.
C	Correct	Smaller steps stop overshooting the minimum.
D	Confusion set	Overfitting shows rising validation loss while training loss falls.

4. Answer C Concept 3

In a small network ~~built for a university research project~~, $y = \sigma(z)$ with $z = w \cdot x$. For one input ~~from a 10,000-row dataset~~, $x = 3$ and $\sigma(z) = 0.8$. What is $\partial y / \partial w$?

Target word: derivative through a sigmoid **Decides:** $\partial y / \partial w = \sigma(1 - \sigma) \cdot x$.

A	Dropped element	Leaves out the $\partial z / \partial w = x$ factor.
B	Adjacent category	Uses σ itself instead of its derivative: 0.8×3 .
C	Correct	$0.8 \times 0.2 \times 3 = 0.48$.
D	Inverted operation	Sign error; the sigmoid's slope is positive.

5. Answer D Concept 4

A churn model at a telecom ~~with 9 million subscribers~~ scores 98% on training data and 81% on validation. The team, ~~which recently moved to a new building~~, increases the L2 penalty λ . What is the most likely result?

Target word: effect of stronger regularization **Decides:** L2 shrinks weights: more bias, less variance.

A	Absolute language	Regularization narrows the gap; it cannot remove all error.
B	Confusion set	Regularization lowers variance.
C	Inverted operation	Penalizing weights limits fit; it does not raise training accuracy.
D	Correct	Regularization trades a bit of fit for better generalization.

6. Answer B Concept 5

A factory's visual inspection model, ~~sold by a vendor with glowing reviews~~, catches 90% of defective parts and falsely flags 5% of good parts. Only 2% of parts are defective. ~~The factory runs three shifts~~. If a part is flagged, about how likely is it truly defective?

Target word: P(defect | flag) **Decides:** Base rate 2%: 0.018 true flags vs. 0.049 false flags.

A	Confusion set	That is P(flag defect), the sensitivity.
B	Correct	$0.9 \times 0.02 = 0.018$; $0.05 \times 0.98 = 0.049$; $0.018 / 0.067 \approx 0.27$.
C	Dropped element	The prior before any flag is seen.
D	Inverted operation	1 – 5% is the rate of correctly passing good parts.

7. Answer D Concept 5 · transfer

Using the same numbers (2% defective, 90% caught, 5% of good parts falsely flagged), what share of all parts gets flagged?

Target word: solve for P(flag) **Decides:** $P(\text{flag}) = 0.9(0.02) + 0.05(0.98)$.

A	Dropped element	Counts only true positives.
B	Confusion set	Counts only false positives.
C	Adjacent category	Adds the two rates ($0.90 + 0.05$) without weighting by base rate.
D	Correct	$0.018 + 0.049 = 0.067$.

8. Answer C Concept 6

A classifier for a wildlife nonprofit ~~with volunteers on three continents~~ assigns probability 0.25 to the correct species and 0.75 to the others combined. ~~The photo was taken at dawn~~. Using the natural log, what is the cross-entropy loss for this example?

Target word: cross-entropy value **Decides:** Loss = $-\ln(0.25)$.

A	Adjacent category	That is $-\log_2 0.25$, in bits, not the natural log.
B	Wrong stem number	The probability itself, not the loss.
C	Correct	$-\ln 0.25 = \ln 4 \approx 1.386$.
D	Inverted operation	$-\ln 0.75$ uses the wrong-class probability.

9. Answer A Concept 6 · transfer

One example's cross-entropy loss (natural log) is 0.105. About what probability did the model give the true class?

Target word: solve for p **Decides:** $p = e^{-0.105}$.

A	Correct	$e^{-0.105} \approx 0.90$.
B	Wrong stem number	Repeats the loss value.
C	Inverted operation	That is $1 - p$, the probability left for other classes.
D	Dropped element	Drops the minus sign; a probability cannot exceed 1.

10. Answer B Concept 7

A 12-layer network using sigmoid activations, ~~trained overnight in a lab with a view of the harbor,~~ shows gradients at the first layer about one-millionth the size of those at the last. ~~The lab just bought new chairs.~~ What is the best fix?

Target word: remedy for vanishing gradients **Decides:** Many sigmoid derivatives (≤ 0.25) multiplied together.

A	Inverted operation	More small factors shrink gradients further.
B	Correct	Both keep gradient products from shrinking layer by layer.
C	Confusion set	They hit the earliest layers hardest.
D	Adjacent category	Would destabilize later layers without fixing the cause.

11. Answer A Concept 8

A fraud model at an online marketplace, ~~which just launched in Brazil,~~ scores 80% accuracy on a test set of 2,500 examples. ~~The test set was labeled by contractors.~~ What is the standard error of that accuracy estimate?

Target word: standard error **Decides:** $SE = \sqrt{(0.8 \times 0.2 / 2,500)}$.

A	Correct	$\sqrt{0.000064} = 0.008$.
B	Confusion set	That is the error rate, $1 - \text{accuracy}$.
C	Dropped element	Leaves out the square root.
D	Adjacent category	That is the 95% margin of error ($1.96 \times SE$), not SE.

12. Answer D Concept 8 · transfer

With 1,000 test examples, the margin of error is ± 3 points. About how many examples are needed for ± 1.5 points?

Target word: solve for n **Decides:** Error scales with $1/\sqrt{n}$; halving it needs $4\times$ the data.

A	Confusion set	Multiplies by $\sqrt{2}$ instead of 2^2 .
B	Inverted operation	Fewer examples would widen the margin.
C	Adjacent category	Assumes error falls in direct proportion to n.
D	Correct	$(3 / 1.5)^2 \times 1,000 = 4,000$.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	7	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	10	
Absolute language	Always, never, guarantee — where judgment applies.	1	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	0	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Gradient descent update	$w_{\text{new}} = w - \eta \cdot \partial L / \partial w$: step against the gradient, scaled by the learning rate.	Adding the gradient, or forgetting the learning rate.	A weight, a gradient, and a learning rate.
2	Learning rate and convergence	Too large a learning rate makes loss oscillate or blow up; too small makes progress very slow.	Raising the learning rate when loss is already oscillating upward.	A loss curve that zigzags or grows epoch to epoch.
3	Chain rule in backpropagation	Multiply local derivatives along the path; for a sigmoid, $d\sigma/dz = \sigma(1 - \sigma)$.	Using the activation value in place of its derivative, or dropping the input factor.	A nested function such as $\sigma(wx)$ and a derivative with respect to w .
4	Regularization and the bias-variance trade-off	Stronger L2 regularization shrinks weights, raising bias slightly and lowering variance, which narrows the train-validation gap.	Expecting regularization to lower training error.	An overfit model and a change to λ or weight decay.
5	Bayes' theorem and base rates	$P(A B) = P(B A)P(A) / P(B)$; with a rare condition, most positives can be false positives.	Confusing sensitivity $P(\text{flag} \text{defect})$ with $P(\text{defect} \text{flag})$.	A small prior, a detection rate, and a false-positive rate.
6	Cross-entropy loss	For one example, cross-entropy = $-\ln(p)$ assigned to the true class; lower confidence in the truth means higher loss.	Using the probability of the wrong class, or mixing log bases.	A predicted probability for the correct label.
7	Vanishing gradients	Backpropagated gradients are products of layer derivatives; many factors below 1 (sigmoid max 0.25) shrink them toward zero, fixed by ReLU-type activations, residual connections, and careful initialization.	Adding more layers or a huge learning rate to fix tiny early-layer gradients.	Early layers barely change while later layers train.
8	Sampling error in evaluation	Standard error of an accuracy estimate is $\sqrt{p(1 - p)/n}$; halving the error needs four times the data.	Assuming doubling the test set halves the error.	A test-set size and a margin of error.

Course scope (public description): This course enables learners to apply advanced mathematics, including calculus, linear algebra, probability, statistics, and optimization, to analyze and solve problems in artificial intelligence engineering. Learners use mathematical reasoning to model, analyze, and evaluate machine learning and deep learning systems, with emphasis on model training dynamics, generalization, stability, and performance. The course develops learners' ability to analyze complex AI engineering problems by applying linear algebra techniques such as vector spaces and matrix factorizations, evaluate gradients, loss surfaces, and convergence behavior using multivariable calculus and optimization, and assess uncertainty and performance through probability, statistics, and information theory. Learners conduct computational experiments and data-driven analysis using Python, NumPy, and appropriate machine learning libraries to validate mathematical models and support engineering decisions. Throughout the course, learners demonstrate the ability to interpret results, draw justified conclusions, and communicate mathematical reasoning in the context of AI system design and evaluation.